Math 110 — Week 11

Solutions

HW11 P1. Idea. The standard basis is orthonormal, so $\mathcal{M}(T^*)$ is the conjugate transpose of $\mathcal{M}(T)$ (Prop. 93). T moves each coordinate one place down; its adjoint moves it one place up.

1. $Te_j = e_{j+1}$ for $j < n$ and $Te_n = 0$, so $\mathcal{M}(T)$ has 1 in each entry $(j+1, j)$ and 0 elsewhere. Its entries are real, so the conjugate transpose has 1 in each entry $(j, j+1)$: $T^*e_{j+1} = e_j$ and $T^*e_1 = 0$.
2. Hence
$$T^*(z_1, \dots, z_n) = (z_2, z_3, \dots, z_n, 0).$$
3. Check with the definition: both sides below equal $z_1\bar{w}_2 + \cdots + z_{n-1}\bar{w}_n$.
$$\langle Tz, w\rangle = \sum_{j=2}^{n} z_{j-1}\bar{w}_j, \qquad \langle z, (w_2, \dots, w_n, 0)\rangle = \sum_{k=1}^{n-1} z_k\bar{w}_{k+1}.$$

HW11 P2. Idea. Prop. 94 turns a statement about $\operatorname{null} T$ into one about $\operatorname{range} T^*$; orthogonal complements turn "$= \{0\}$" into "$= V$".

1. T is injective exactly when $\operatorname{null} T = \{0\}$.
2. By (T5), $\operatorname{null} T = \{0\}$ exactly when $(\operatorname{null} T)^\perp = V$: if $\operatorname{null} T = \{0\}$, its complement has dimension n; conversely, if $(\operatorname{null} T)^\perp = V$, then $\operatorname{null} T = ((\operatorname{null} T)^\perp)^\perp = V^\perp = \{0\}$.
3. By Prop. 94(ii), $(\operatorname{null} T)^\perp = \operatorname{range} T^*$. So T is injective exactly when $\operatorname{range} T^* = V$, that is, when T^* is surjective.

HW11 P3. Idea. In the standard basis $\mathcal{M}(T)$ is a real matrix, so T^* is its transpose; the rest is multiplication and the characteristic polynomial.

1. **(a).** $T(1,0) = (0,1)$ and $T(0,1) = (-1,0)$, so $\mathcal{M}(T)$ and its transpose $\mathcal{M}(T^*)$ are
$$\mathcal{M}(T) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \qquad \mathcal{M}(T^*) = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \qquad T^*(x,y) = (y,-x).$$
2. **(b).** $TT^*(x,y) = T(y,-x) = (x,y)$ and $T^*T(x,y) = T^*(-y,x) = (x,y)$, so $TT^* = I = T^*T$: T is normal. $T^* = -T \neq T$ (for instance $T(1,0) = (0,1)$ but $T^*(1,0) = (0,-1)$), so T is not self-adjoint.
3. **(c).** The characteristic polynomial is $\lambda^2 - (\operatorname{tr})\lambda + \det = \lambda^2 + 1$. Over $\mathbb{R}$ it has no root, so T has no eigenvalue. Over $\mathbb{C}$ the eigenvalues are i and $-i$, with eigenvectors $(1,-i)$ and $(1,i)$: $T(1,-i) = (i,1) = i(1,-i)$ and $T(1,i) = (-i,1) = -i(1,i)$.

HW11 P4. Idea. A projector splits $V = \operatorname{range} P \oplus \operatorname{null} P$ (Week 10, Prop. 91). It is the orthogonal projection exactly when the two pieces are perpendicular, and normality gives $\operatorname{null} P = (\operatorname{range} P)^\perp$ through Prop. 99.

1. **(a)⇒(b).** A self-adjoint operator is normal: $PP^* = P^2 = P^*P$.
2. **(b)⇒(c).** By Prop. 99, $(\operatorname{null} P)^\perp = \operatorname{range} P$; taking complements (T5), $\operatorname{null} P = (\operatorname{range} P)^\perp$. Since $P^2 = P$, $V = \operatorname{range} P \oplus \operatorname{null} P$, P is the identity on $\operatorname{range} P$ (if $u = Pv$ then $Pu = P^2v = u$) and 0 on $\operatorname{null} P$. So P fixes $\operatorname{range} P$ and sends $(\operatorname{range} P)^\perp$ to 0: it is the orthogonal projection onto $\operatorname{range} P$.

HW11 P4, continued.

3. **(c)⇒(a).** Let $U = \operatorname{range} P$, and write $v = u + u'$, $w = x + x'$ with $u, x \in U$ and $u', x' \in U^\perp$ (T5). Then $Pv = u$ and $Pw = x$, so
$$\langle Pv, w\rangle = \langle u, x + x'\rangle = \langle u, x\rangle, \qquad \langle v, Pw\rangle = \langle u + u', x\rangle = \langle u, x\rangle,$$
since $\langle u, x'\rangle = 0$ and $\langle u', x\rangle = 0$. So $\langle Pv, w\rangle = \langle v, Pw\rangle$ for all v, w: $P^* = P$ (T4).

HW11 P5. Idea. To show $Tv \perp U$, move T to the other slot, where it acts on a vector of U and stays in U.

1. Let $v \in U^\perp$ and $u \in U$. Since U is T-invariant, $Tu \in U$, so $\langle v, Tu\rangle = 0$.
2. Since T is self-adjoint, $\langle Tv, u\rangle = \langle v, Tu\rangle = 0$. This holds for every $u \in U$, so $Tv \in U^\perp$.

HW11 P6. Idea. A is symmetric with distinct eigenvalues, so its eigenvectors are automatically orthogonal; normalize them and use $A^m = QD^mQ^\mathsf{T}$.

1. **(a).** $\operatorname{tr} A = 4$ and $\det A = 3$, so $\lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1)$. $A(1, 1) = (3, 3)$ and $A(1, -1) = (1, -1)$, so $(1, 1)$ is an eigenvector for 3 and $(1, -1)$ for 1. They are orthogonal; normalized, $u_1 = \frac{1}{\sqrt{2}}(1, 1)$, $u_2 = \frac{1}{\sqrt{2}}(1, -1)$.
2. **(b).** With these columns,
$$Q = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad D = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}, \qquad A = QDQ^\mathsf{T}.$$
3. **(c).** $A^m = QD^mQ^\mathsf{T}$, and multiplying out,
$$A^m = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 3^m & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 3^m + 1 & 3^m - 1 \\ 3^m - 1 & 3^m + 1 \end{bmatrix}.$$
For $m = 1$ this gives back A.

HW11 P7. Idea. Use both directions of the complex spectral theorem: an orthonormal eigenbasis of T is also one of $p(T)$.

1. By Prop. 104, V has an orthonormal basis $e_1, \ldots, e_n$ with $Te_j = \lambda_j e_j$.
2. Then $T^k e_j = \lambda_j^k e_j$ for every k, so $p(T)e_j = p(\lambda_j)e_j$: the same orthonormal basis consists of eigenvectors of $p(T)$.
3. By Prop. 104 again (the other direction), $p(T)$ is normal.

O1. Idea. The standard basis is not orthonormal for this inner product ($\|e_1\|^2 = 2$), so solve the defining equation directly.

1. For $x, y \in \mathbb{R}^2$, $\langle Tx, y\rangle = \langle (x_2, 0), y\rangle = 2x_2y_1$. Write $T^*y = (p, q)$; then $\langle x, T^*y\rangle = 2x_1p + x_2q$.
2. These agree for every x exactly when $2p = 0$ and $q = 2y_1$. So $T^*(y_1, y_2) = (0, 2y_1)$.
3. The matrices in the standard basis are below: $\mathcal{M}(T^*)$ is not the transpose of $\mathcal{M}(T)$.
$$\mathcal{M}(T) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad \mathcal{M}(T^*) = \begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix}, \qquad \mathcal{M}(T)^\mathsf{T} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

O2. Idea. $E(\lambda, T) = \operatorname{null}(T - \lambda I)$ and $E(\bar\lambda, T^*) = \operatorname{null}(T^* - \bar\lambda I) = \operatorname{null}\big((T - \lambda I)^*\big)$, and LADR 7.7 compares the null space of a map with that of its adjoint.

1. By LADR 7.5, $(T - \lambda I)^* = T^* - \bar\lambda I$, so $E(\bar\lambda, T^*) = \operatorname{null}\big((T - \lambda I)^*\big)$.
2. LADR 7.7 for $S = T - \lambda I \in \mathcal{L}(V, V)$ gives $\dim \operatorname{null} S^* = \dim \operatorname{null} S + \dim V - \dim V = \dim \operatorname{null} S$.
3. So $\dim E(\bar\lambda, T^*) = \dim E(\lambda, T)$. (When λ is not an eigenvalue, both sides are 0.)

O3. Idea. Compare M with M^*, then compare MM^* with M^*M; if normal, the eigenvectors for the two distinct eigenvalues are orthogonal.

1. M^* (conjugate transpose) differs from M, so M is not self-adjoint; but $MM^* = M^*M = 2I$, so M is normal:
$$M^* = \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix}, \qquad MM^* = \begin{bmatrix} 1+1 & -i+i \\ i-i & 1+1 \end{bmatrix} = 2I = M^*M.$$
2. $\operatorname{tr} M = 2$ and $\det M = 1 - i^2 = 2$, so the eigenvalues are the roots of $\lambda^2 - 2\lambda + 2$: $1 \pm i$. $M(1,1) = (1+i)(1,1)$ and $M(1,-1) = (1-i,-1+i) = (1-i)(1,-1)$.
3. An orthonormal eigenbasis (Prop. 104): $\frac{1}{\sqrt{2}}(1,1)$, $\frac{1}{\sqrt{2}}(1,-1)$. The eigenvalues are not real, consistent with M not being self-adjoint.

O4. Idea. One direction is Prop. 95. For the other, write T in an orthonormal eigenbasis: a real diagonal matrix is its own conjugate transpose.

1. **Assume T is self-adjoint.** Its eigenvalues are real by Prop. 95.
2. **Assume every eigenvalue is real.** By Prop. 104 there is an orthonormal basis in which $\mathcal{M}(T)$ is diagonal with entries $\lambda_1, \dots, \lambda_n \in \mathbb{R}$.
3. By Prop. 93, $\mathcal{M}(T^*)$ is the conjugate transpose, which is the same real diagonal matrix. So $T^* = T$.

O5. Idea. On an orthonormal eigenbasis, $T^9 = T^8$ becomes $\lambda^9 = \lambda^8$ for each eigenvalue, which forces $\lambda \in \{0, 1\}$.

1. By Prop. 104, V has an orthonormal basis $e_1, \dots, e_n$ with $Te_j = \lambda_j e_j$.
2. Applying $T^9 = T^8$ to e_j: $\lambda_j^9 e_j = \lambda_j^8 e_j$, so $\lambda_j^8(\lambda_j - 1) = 0$ and $\lambda_j \in \{0, 1\}$.
3. All eigenvalues are real, so T is self-adjoint by O4.
4. $T^2 e_j = \lambda_j^2 e_j = \lambda_j e_j = Te_j$ for each j (as $0^2 = 0$, $1^2 = 1$), so $T^2 = T$ on a basis, hence everywhere.

O6. Idea. For self-adjoint T, $\|Tv\|^2 = \langle T^2 v, v\rangle$.

1. For every v, $\|Tv\|^2 = \langle Tv, Tv\rangle = \langle T^*Tv, v\rangle = \langle T^2 v, v\rangle = 0$, using $T^* = T$ and $T^2 = 0$.
2. So $Tv = 0$ for every v: $T = 0$.